Auditing AI-Based ESG Scores: Transparency, Bias, Robustness, and Sustaiability Claims

AI, Power & Responsibility – Governing Intelligent Systems for Sustainability

SCAN FOR CODE & DATA

1 INTRODUCTION

AI-based ESG scoring tools are increasingly used by investors and regulators. However, concerns remain about transparency, fairness, robustness, and potential bias. This study evaluates AI-generated ESG risk scores using two public datasets and multiple machine learning models.

2 DATASETS

Dataset 1: Public Company ESG Risk Ratings (Kaggle)
- ~7,000 companies
- ESG risk scores: Environment, Social, Governance, Controversy
- Target: Total ESG Risk score

Dataset 2: S&P 500 ESG Sustainability Reports (Kaggle)
- ~5,000 reports
- Text + ESG scores

3 METHODOLOGY PIPELINE

Data Collection
Compile ESG datasets

Preprocessing
Handle missing values, clean & encode data

Model Training
Train ML models on structured ESG features

Evaluation & Audit
Evaluate models using RQ1–RQ5

Insights & Recommendations
Derive insights and propose responsible AI guidelines

RESEARCH QUESTIONS (RQ)

- RQ1 Which ML model predicts ESG risk scores most accurately?
- RQ2 Which ESG factors most influence AI-generated ESG scores?
- RQ3 How robust are ESG predictions when sustainability data is missing?
- RQ4 How sensitive are ESG predictions to noisy sustainability data?
- RQ5 Does the AI model exhibit industry-level bias?

DATASET (For Modeling)

Source: SP 500 ESG Risk Ratings Dataset (Kaggle)

Features Used:
- Environment Risk Score
- Social Risk Score
- Governance Risk Score
- Controversy Score

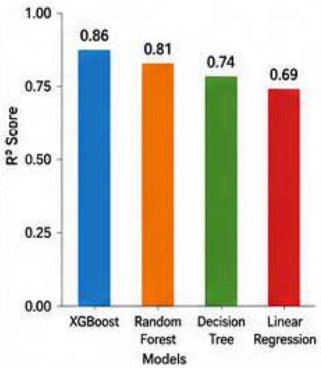
Target:
- Total ESG Risk score

METHODOLOGY PIPELINE

Data Collection → Preprocessing → Model Training (ML Models) → Evaluation & Audit (RQ1–RQ5) → Insights & Recommendations

RQ1 Which ML Model Predicts ESG Scores Most Accurately?

Model Performance Comparison (R^2 Score)

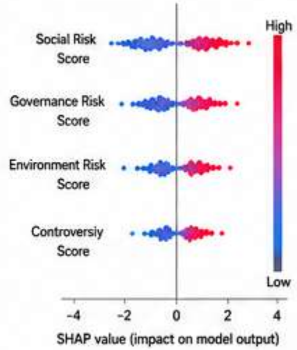


KEY TAKEAWAY

XGBoost achieved the highest R^2 score (0.86), demonstrating superior predictive performance for ESG risk scores.

RQ2 Which ESG Factors Most Influence AI-Generated ESG Scores?

SHAP Summary Plot

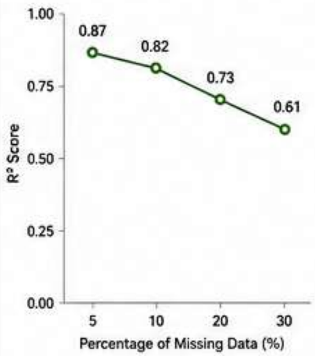


KEY TAKEAWAY

Social Risk Score has the greatest influence on ESG risk predictions, followed by Governance, Environment, and Controversy Score.

RQ3 How Robust Are ESG Predictions When Sustainability Data Is Missing?

Missing Data (%) vs R^2 Score

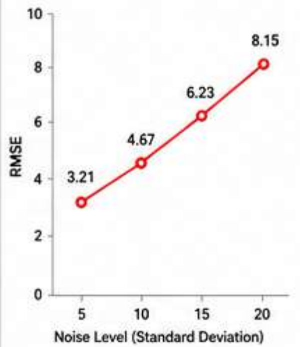


KEY TAKEAWAY

Model performance gradually declines as missing data increases, but remains reasonably robust up to 20% missing data.

RQ4 How Sensitive Are ESG Predictions to Noisy Sustainability Data?

Noise Level vs RMSE

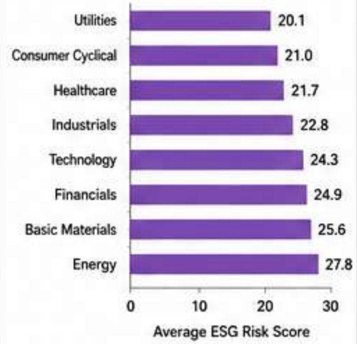


KEY TAKEAWAY

Prediction error (RMSE) increases consistently with higher noise levels, indicating sensitivity to noisy sustainability data.

RQ5 Does the AI Model Exhibit Industry-Level Bias?

Average ESG Risk Score by Sector



KEY TAKEAWAY

Significant variation exists across sectors. Energy shows the highest ESG risk, indicating potential industry-level bias.

KEY CONTRIBUTIONS

- Explainable AI for ESG scoring using SHAP.
- Fairness audit across industry sectors.
- Robustness under missing and noisy data.
- Comprehensive model comparison.
- AI-driven recommendations for transparent & responsible ESG scoring.

CONCLUSION

Our audit framework provides a holistic view of AI-based ESG scoring systems. While models are effective, they are sensitive to data quality and show industry-level variations. Integrating explainability, robustness checks, and fairness audits is essential for building trustworthy, transparent, and accountable ESG intelligence systems.

KEYWORDS

ESG • Explainability • Fairness • Robustness
Greenwashing • AI Governance • Sustainability